CAPSTONE PROJECT REPORT

AI DATASHEET ASSISTANT

A Retrieval-Augmented Question Answering and Verification Workflow for Operational Amplifiers

Programme	Electronics and Communication Engineering (ECE)
Project domain	Artificial Intelligence for Electronic Component Engineering
Selected component family	Operational Amplifiers (Op-Amps)
Prepared for	ECE Capstone Project
Student name	_____
Register number	_____
Institution / Department	_____
Academic year	2026–2027

Project scope: collect manufacturer datasheets, build a source-grounded assistant, test 30 engineering questions, verify every numerical answer against the original document, and report limitations honestly.

Submitted in partial fulfilment of the requirements for the ECE capstone project

ABSTRACT

Electronic engineers frequently consult long component datasheets to identify supply limits, input ranges, bandwidth, slew rate, package information and electrical characteristics. Manual searching is time-consuming, while a general-purpose AI model may confuse device variants, omit test conditions, mix typical and maximum values, or invent specifications. This project proposes an AI Datasheet Assistant for operational amplifiers. It combines manufacturer-source documents, document retrieval, structured question answering, page-level citations and a human verification step. The assistant is designed to answer direct lookups, comparisons and calculations, and to refuse questions that cannot be supported by the loaded documents. A 30-question test suite covers lookup, comparison, calculation, unanswerable and ambiguous cases. The report provides a reproducible workflow, sample calculations, prompt templates, a verification register, automation design, quick-reference card and presentation outline. The numerical examples are explicitly tied to device variants and datasheet conditions. The project is a prototype specification and demonstration report; actual live model benchmark scores must be filled from the execution log rather than assumed.

Keywords: operational amplifier, datasheet, retrieval-augmented generation, hallucination, citation, verification, ECE.

DECLARATION / NOTE ON EVIDENCE

This report distinguishes (1) manufacturer-published specifications, (2) calculations derived from stated circuit assumptions, and (3) proposed test procedures. A source URL is not a substitute for downloading and checking the exact PDF revision. Before final institutional submission, the student should download the listed PDFs, record revision/date and page numbers, run the 30 tests in the selected AI tools, and attach screenshots or exported logs. No unexecuted model benchmark is represented as a measured result.

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1. INTRODUCTION AND PROBLEM STATEMENT

A semiconductor datasheet is the primary technical reference for using an integrated circuit correctly. It contains absolute maximum ratings, recommended operating conditions, electrical characteristics, typical performance graphs, pin configuration, package details and application circuits. These sections are not interchangeable. For example, an absolute maximum rating is a stress limit and must not be treated as a recommended operating point. Likewise, a typical value is not a guaranteed maximum or minimum.

Problem statement

How can an AI assistant help ECE engineers retrieve accurate operational-amplifier specifications from multiple manufacturer datasheets, show the exact evidence, perform supported calculations, and avoid returning unsupported or incorrectly interpreted values?

Proposed solution

The proposed assistant indexes manufacturer datasheets, retrieves relevant passages for each question, asks a language model to answer only from those passages, and attaches document/page/section evidence. A verification layer checks units, device suffixes, test conditions and arithmetic. If evidence is absent or conflicting, the assistant responds that the answer is not established by the loaded documents and requests clarification.

2. OBJECTIVES AND SCOPE

- Collect at least ten manufacturer datasheets for operational amplifiers from official manufacturer sites.
- Create a searchable document collection with device name, manufacturer, revision, date and source URL metadata.
- Prepare 30 test questions across direct lookup, comparison, calculation, unanswerable and ambiguous categories.
- Load the documents into at least two document-Q&A; systems and compare their answers and citations.
- Use at least four AI tools across document Q&A;, prompting, research, analysis and presentation/automation.
- Manually verify all numerical answers against the original PDF, including units and test conditions.
- Measure answer accuracy, citation correctness, wrong-unit errors, unsupported claims and abstention behaviour.
- Produce a one-page quick-reference card, a 5–7 slide presentation, prompt library and final demonstration.

Scope exclusions: this prototype does not replace engineering review, SPICE simulation, laboratory testing, safety qualification or manufacturer support. It does not certify that a component is suitable for a circuit.

3. BACKGROUND AND SYSTEM REQUIREMENTS

Important datasheet concepts

Term	Meaning / interpretation
Supply voltage	State whether the figure is total supply (V+ – V–), single-supply voltage, or ± dual rails.
GBW	Gain-bandwidth product; do not assume it equals closed-loop bandwidth for every circuit.
Slew rate	Maximum output-voltage rate of change, normally V/µs; large-signal performance can be slew-limited.
Input offset voltage	Differential input voltage needed to make output zero under specified conditions.
Input bias current	Current into or out of each input; note maximum/typical and temperature conditions.
Common-mode range	Allowed voltage range shared by the two inputs; it is not automatically rail-to-rail.
Output swing	Output range under stated supply, load and temperature; not necessarily reaching either rail.
Absolute maximum rating	Stress boundary. It is not a recommended operating condition.

Functional requirements

- Accept natural-language questions and identify the target part and variant.
- Retrieve relevant chunks from the correct manufacturer datasheet.

- Answer with value, unit, conditions, page/section and source document.
- Show calculation steps for derived values and state assumptions.
- Return an explicit 'not found in supplied sources' result for unsupported questions.
- Allow the user to inspect the evidence passage and open the original PDF.

Non-functional requirements: traceability, repeatability, low latency, secure document handling, clear uncertainty, and no hidden substitution between part variants.

4. DATASHEET COLLECTION AND SOURCE REGISTER

The following ten official manufacturer product/document pages form the initial collection. Download the PDF from each page, retain the manufacturer filename, and record the PDF revision printed on its title page. Product pages may offer more than one grade or datasheet revision; the exact selected file must be recorded in the project dataset manifest.

No.	Part / family	Manufacturer	Official source
1	LM358 / LM358B	Texas Instruments	https://www.ti.com/lit/ds/symlink/lm358a.pdf Dual general-purpose amplifier family; use the exact B/BA suffix and revision.
2	LM741	Texas Instruments	https://www.ti.com/product/LM741 Single general-purpose bipolar op amp; datasheet linked from product page.
3	OP07	Texas Instruments	https://www.ti.com/product/OP07 Precision, low-offset single op amp; verify exact grade and conditions.
4	TL081	STMicroelectronics	https://www.st.com/en/amplifiers-and-comparators/tl081.html JFET-input single op amp.
5	TL071	STMicroelectronics	https://www.st.com/en/amplifiers-and-comparators/tl071.html JFET-input single op amp.
6	LM324	STMicroelectronics	https://www.st.com/content/st_com/en/products/amplifiers-and-comparators/operational-amplifiers-op-amps/standard-op-amps/lm324.html Quad general-purpose amplifier family.
7	MCP6002	Microchip Technology	https://www.microchip.com/en-us/product/MCP6002 Dual, low-power 1 MHz op amp family.
8	NE5532	Texas Instruments	https://www.ti.com/product/NE5532 Dual low-noise audio op amp.
9	OPA2134	Texas Instruments	https://www.ti.com/product/OPA2134 Dual audio op amp; FET input.
10	AD822	Analog Devices	https://www.analog.com/en/products/ad822.html Dual, single-supply FET-input amplifier.

Dataset manifest fields

For every downloaded PDF, create one record with: filename; manufacturer; exact orderable part number(s); document title; document number; revision; publication/revision date; download date; source URL; page count; SHA-256 checksum; and notes on variants covered. Keep superseded revisions separate and never merge specifications across revisions without an explicit comparison.

Recommended folder layout

```
project/
datasheets/original_pdfs/
datasheets/manifest.csv
extracted_text/
questions/test_set_30.csv
results/tool_A/
results/tool_B/
verification/answer_audit.csv
prompts/prompt_library.md
presentation/
```

Important: the source register is a collection plan grounded in official manufacturer pages. The student must save the actual PDFs and confirm the revision/page count before claiming that the collection is complete.

5. SYSTEM ARCHITECTURE AND METHODOLOGY

Architecture

User question → query parser → document retriever → relevant datasheet passages → LLM answer generator → citation formatter → verification checks → final answer / abstention.

Module	Function	Output
Document ingestion	Download PDFs, extract text, preserve page numbers and tables.	Page-aware text and metadata.
Chunking	Split by heading/section; retain page, part number and revision metadata.	Searchable chunks.
Embedding / index	Create vector or hybrid lexical index.	Ranked evidence passages.
Query understanding	Identify part, variant, requested parameter and task type.	Structured query.
Answer generation	Use only retrieved evidence; quote value and test conditions.	Draft answer with evidence.
Verification	Check source, units, suffix, min/typ/max, arithmetic and citations.	Pass / flag / abstain.
Logging	Store question, answer, evidence, tool/model and audit verdict.	Reproducible test record.

Recommended implementation sequence

- Acquire and hash the ten original manufacturer PDFs.
- Extract text with a PDF parser or document-Q&A; service; inspect tables and scanned pages visually when extraction is poor.
- Chunk by datasheet headings rather than arbitrary large blocks. Keep a small overlap and never detach a table value from its header/conditions.
- Store metadata on every chunk: manufacturer, part family, exact variant, revision, page, section and source filename.
- Retrieve the top passages using a hybrid of keyword and semantic search. Filter by exact part/variant first.
- Generate a concise answer with a fixed evidence schema. If evidence does not contain the requested fact, abstain.
- Run deterministic checks: units, numeric parsing, formula calculation, source page and variant match.
- Export the conversation and audit verdict to CSV/Google Sheets.

Answer record schema

```
question_id, question, part_number, answer, numeric_value, unit, min_typ_max, conditions, document,
revision, page, section, evidence_quote, confidence, verification_status, reviewer_notes
```

Two document-Q&A; platforms can be compared (for example, NotebookLM and ChatGPT with uploaded PDFs). Other supporting tools can include Gemini, Claude, Perplexity, Google Sheets, Python/Colab, Canva and LTspice. Tool availability and feature names can change; record the actual product/version and date used.

4+ AI/tool mapping

Tool	Project role	Evidence to retain
NotebookLM	Load source PDFs; ask source-grounded questions.	Notebook/source list, answer screenshots and citations.
ChatGPT	Structured Q&A;, prompt testing, calculations and report drafting.	Prompt, model/date, transcript and audit.
Gemini or Claude	Independent second-model comparison.	Same question set and source files.
Perplexity	Locate official manufacturer pages and current documents.	Search results and final official URL checks.
Google Sheets / Colab	Log answers, calculate metrics and produce plots.	CSV, formulas, notebook.

Tool	Project role	Evidence to retain
Canva / PowerPoint	Create final 5–7 slide presentation.	Exported slides.

6. PROMPT ENGINEERING AND ANSWER FORMAT

Baseline prompt (plain question)

What is the gain-bandwidth product of the LM358?

This prompt is easy to write but may produce an uncited answer, confuse variants, or omit the test conditions and datasheet revision.

Role + evidence-constrained prompt

You are an electronics datasheet analyst. Answer the question using only the attached manufacturer datasheet(s). First identify the exact part number, suffix and document revision. Return: (1) direct answer, (2) value and unit, (3) whether it is minimum, typical or maximum, (4) operating/test conditions, (5) datasheet page and section, and (6) a short supporting quotation. Do not infer missing values. If the supplied document does not explicitly support the answer, say “Not found in the supplied datasheet.” If multiple variants disagree, show each separately and ask which variant is intended. For calculations, show formula, substitutions and units. Never treat absolute maximum ratings as recommended operating conditions.

Structured output prompt

Return a table with columns: Part number | Parameter | Value | Unit | Min/Typ/Max | Conditions | Document revision | Page/section | Evidence | Verification status. Do not fill a cell by guessing. Use “Not stated” when evidence is absent.

Comparison prompt

Compare LM358B and LM741 for supply voltage, gain-bandwidth product, slew rate, input offset voltage and output swing. Use only the attached manufacturer datasheets. Keep the exact variant and test conditions visible. Do not rank one as universally better; explain which parameter matters for which application. Mark any non-comparable condition as “not directly comparable.”

Ambiguity-resolution prompt

The part number is incomplete. Ask one clarification question before answering. Do not silently assume a suffix, package, grade or manufacturer. If the user permits a family-level answer, clearly label the family-level range and its limitations.

Prompt experiment design

- Prompt A: plain question, no role and no answer schema.
- Prompt B: role + quote-the-section + cite-page prompt.
- Prompt C: strict evidence-only prompt with explicit abstention and variant checks.
- Run each prompt on the same test question, same PDF set and same model version; randomize order if possible.
- Compare correctness, citation quality, unit errors, unsupported claims, answer completeness and response time.

7. THIRTY-QUESTION TEST SUITE

Use the following fixed test set for every AI platform and prompt condition. Preserve the question wording so comparisons are fair. The expected behaviour column is the grading key, not a pre-filled AI result.

ID	Category	Test question	Expected evidence / behaviour
Q01	Direct lookup	What is the typical gain-bandwidth product listed for the selected LM358 variant?	Find GBW row; state variant, revision, conditions and page.
Q02	Direct lookup	What is the typical slew rate of the LM741?	Give V/µs and test condition.
Q03	Direct lookup	How many independent amplifiers are inside an LM324?	Confirm device family and package variant.
Q04	Direct lookup	What input technology is used by TL081?	Identify JFET input from description.
Q05	Direct lookup	What is the typical gain-bandwidth product of OPA2134?	Use exact OPA2134 datasheet row.
Q06	Direct lookup	What is the supply-voltage range of AD822?	Distinguish single supply from dual supply.
Q07	Direct lookup	How many channels does MCP6002 contain?	Use Microchip document, not a reseller.
Q08	Direct lookup	What is the typical slew rate of NE5532?	State conditions and units.
Q09	Direct lookup	What is the maximum input offset voltage for OP07 grade chosen?	Specify grade and temperature range.
Q10	Direct lookup	Does TL071 have internal frequency compensation?	Cite feature/description section.
Q11	Comparison	Compare LM358 and LM741 supply voltage limits.	Use matching definitions; do not mix total and ± rails.
Q12	Comparison	Compare TL071 and OPA2134 input stage technologies.	Identify JFET/FET input and avoid assuming identical performance.
Q13	Comparison	Which of LM358 and NE5532 has higher typical GBW?	Compare datasheet typical values and variant.
Q14	Comparison	Compare OPA2134 and AD822 typical slew rates.	State each condition; do not overgeneralize.
Q15	Comparison	Compare LM324 and MCP6002 channel counts.	Quad versus dual.
Q16	Calculation	For a non-inverting amplifier with Rf=90 kΩ and Rg=10 kΩ, calculate closed-loop gain.	Use Av=1+Rf/Rg; result 10 V/V.
Q17	Calculation	For an inverting amplifier with Rin=10 kΩ and Rf=47 kΩ, calculate gain.	Use Av=−Rf/Rin; result −4.7 V/V.
Q18	Calculation	A 2 Vpp sine wave at 100 kHz is required. What minimum slew rate is required?	SRmin=2πfVp; Vp=1 V; result ≈0.628 V/µs.
Q19	Calculation	If GBW is 1 MHz, estimate closed-loop bandwidth at noise gain 10.	First-order estimate 100 kHz; state assumptions.
Q20	Calculation	An op amp has 5 mV input offset and closed-loop noise gain 11. Estimate output offset due to input offset alone.	Vout offset ≈55 mV; other errors ignored.
Q21	Calculation	For a 5 V single supply, is an output of 5.0 V guaranteed if the device is not rail-to-rail?	Cannot conclude without output swing/load data; likely no guarantee.
Q22	Unanswerable	What is the exact price of this IC at a local Chennai shop today?	Not in datasheet; requires live seller quote.
Q23	Unanswerable	What is the lifetime of an IC in my circuit under unknown thermal conditions?	Cannot determine from datasheet alone; need application and reliability data.
Q24	Unanswerable	What is the exact noise of my assembled PCB?	Requires circuit/layout/measurement details.
Q25	Unanswerable	Does this device guarantee stability with every capacitive load?	Only answer if datasheet explicitly specifies; otherwise not established.
Q26	Unanswerable	What is the guaranteed output swing at 2 mA load if the datasheet gives no such test point?	Abstain; do not interpolate as guaranteed.
Q27	Ambiguous	What is the maximum voltage of the 358?	Ask LM358 variant and whether supply, input common-mode or differential voltage is meant.

ID	Category	Test question	Expected evidence / behaviour
Q28	Ambiguous	What is the offset of the OP07?	Ask grade, temperature range and whether typical or maximum.
Q29	Ambiguous	Can I use the 741 at 5 V?	Ask single/dual supply and circuit conditions; check minimum total supply.
Q30	Ambiguous	Which op amp is best?	Ask application, supply, bandwidth, noise, load, accuracy, cost and package.

8. WORKED EXAMPLES AND VERIFICATION

Example A — direct datasheet lookup

A manufacturer page for the LM358B family reports a typical gain-bandwidth product of 1.2 MHz and a typical slew rate of 0.5 V/µs under the stated product/datasheet conditions. The exact PDF revision and test row must be checked before using these as a design guarantee. Do not transfer these figures automatically to every LM358-compatible device sold by another manufacturer.

Example B — non-inverting amplifier calculation

Given Rf = 90 kΩ and Rg = 10 kΩ:
Av = 1 + (Rf/Rg) = 1 + (90/10) = 10 V/V.
For an ideal op amp and an input of 0.2 V, the ideal output is 2.0 V. Actual output must remain within the selected device's output swing and linear operating range.

Example C — slew-rate requirement

For a sine wave, SRmin = 2πfVp. With Vpp = 2 V, Vp = 1 V and f = 100 kHz:
SRmin = 2π × 100,000 × 1 = 628,319 V/s ≈ 0.628 V/µs.
A practical design should allow margin and verify distortion, load and temperature conditions.

Example D — bandwidth estimate

For a dominant-pole voltage-feedback amplifier, a first-order estimate is closed-loop bandwidth ≈ GBW / noise gain. If GBW = 1 MHz and noise gain = 10, the estimate is 100 kHz. This is an approximation; verify the datasheet's gain/phase plots and stability requirements for the selected circuit.

Manual verification checklist

- Does the evidence come from the correct manufacturer and exact part number?
- Does the document revision match the manifest?
- Is the number in the correct row and column (minimum / typical / maximum)?
- Are supply voltage, load, temperature and frequency conditions preserved?
- Are units and prefixes correct (mV versus V, nA versus µA, MHz versus kHz)?
- Does the cited page contain the evidence, including tables and footnotes?
- Are calculations dimensionally correct and based on stated assumptions?
- Is the answer appropriately qualified, or should the system abstain?

Common error patterns

Error	Example	Prevention
Variant mix-up	LM358B value applied to LM358 from another vendor.	Filter retrieval by manufacturer and suffix.
Unit error	0.5 V/µs written as 0.5 V/s.	Unit parser and human audit.
Min/typ/max confusion	Typical value described as guaranteed maximum.	Require statistic label in output.
Condition omission	Output swing quoted without load current.	Include test condition field.
Rating misuse	Absolute maximum supply used as operating range.	Separate ratings from recommended conditions.
False citation	Page number does not contain claimed figure.	Open PDF and inspect exact page/table.

9. EVALUATION METRICS AND RESULT SHEET

Do not invent benchmark results. Run the same 30 questions in at least two tools and record the actual output. Grade each response against the original PDF and the expected behaviour in Section 7. A question is fully correct only when the value, unit, variant, statistic type, conditions and citation are all correct where applicable.

Metrics

Metric	Calculation
Answer accuracy	Correct supported answers ÷ answerable questions × 100%.
Citation precision	Citations that directly support the claim ÷ all citations × 100%.
Unit error rate	Answers with incorrect/missing units ÷ numeric answers × 100%.
Hallucination rate	Unsupported factual claims ÷ all factual claims × 100%.
Abstention accuracy	Correct abstentions on unanswerable/ambiguous cases ÷ such cases × 100%.
Completeness	Responses containing all required fields ÷ questions requiring those fields × 100%.

Per-question audit sheet (copy into spreadsheet)

Q ID	Tool / prompt	Correct?	Citation correct?	Unit / condition correct?	Abstained appropriately?	Reviewer note
Q01	_____	Y / N	Y / N	Y / N	Y / N	_____
Q02	_____	Y / N	Y / N	Y / N	Y / N	_____
Q03	_____	Y / N	Y / N	Y / N	Y / N	_____
…	Repeat through Q30	—	—	—	—	_____

Summary result table

System	Prompt	Accuracy	Citation precision	Unit error rate	Hallucination rate	Notes
Tool A	Plain	____%	____%	____%	____%	_____
Tool A	Evidence prompt	____%	____%	____%	____%	_____
Tool B	Plain	____%	____%	____%	____%	_____
Tool B	Evidence prompt	____%	____%	____%	____%	_____

Interpretation: a good system should improve citation precision and reduce unsupported claims under the evidence-constrained prompt. If answer accuracy rises but citation precision falls, the result is not acceptable for engineering use. Report failures, not only successful examples.

10. AUTOMATION AND IMPLEMENTATION PLAN

Low-code workflow

1. User enters question in a Google Form or a simple chat interface.
2. Workflow sends the question and retrieved evidence to the selected AI model.
3. Model returns structured JSON or a fixed table.
4. Automation extracts the fields and appends a row to Google Sheets.
5. Reviewer selects PASS / FAIL / NEEDS REVIEW and enters a note.
6. A dashboard summarizes accuracy, citation and error counts.

Example pseudocode

```
for question in test_questions:
part = identify_part_and_variant(question)
evidence = retrieve(part, question)
if evidence is empty:
answer = abstain("Not found in supplied datasheet")
else:
answer = llm_answer(question, evidence, strict_prompt)
checks = validate_units_variant_conditions(answer, evidence)
log(question, answer, evidence, checks)
reviewer_verdict = human_review(answer, evidence)
append_to_sheet(reviewer_verdict)
```

Implementation options

- Google Sheets: central answer log and formula-based scoring.
- Google Apps Script or Make/Zapier: form-to-sheet and notification automation.
- Python/Colab: PDF parsing, retrieval prototype, unit checks and evaluation metrics.
- NotebookLM / ChatGPT / Gemini / Claude: document-grounded Q&A; comparison.
- LTspice: simulate selected amplifier circuits after datasheet constraints are understood.

Suggested schedule

Week	Task	Deliverable
1	Select variants; download ten official PDFs; complete manifest.	PDF collection + manifest.
2	Extract/index documents; build prompt templates.	Searchable source set.
3	Run 30 questions on two tools and two prompt styles.	Raw transcripts/logs.
4	Manually audit numerical claims and citations.	Verification report.
5	Build automation, quick-reference card and slides.	Automation + presentation.
6	Final demonstration, revisions and viva preparation.	Final report and demo.

11. ONE-PAGE QUICK-REFERENCE CARD

AI DATASHEET ASSISTANT — ENGINEER'S CHECKLIST

Before asking	Specify exact part number, suffix, manufacturer and datasheet revision.
Ask precisely	Name the parameter and whether you need min / typ / max, temperature and load conditions.
Demand evidence	Require page, section, table/figure and a short supporting excerpt.
Check the number	Verify prefix, unit, sign, decimal point and test condition in the PDF.
Check the rating	Never treat absolute maximum ratings as recommended operating conditions.
For calculations	Write formula, substitution, units and assumptions; then check device limits.
If evidence is missing	Use “not stated in supplied datasheet”; do not guess or use a random web snippet.

Before asking	Specify exact part number, suffix, manufacturer and datasheet revision.
Before design release	Review the original PDF, simulate where appropriate, and test the physical circuit.

Core equations

Non-inverting gain: Av = 1 + Rf/Rg
Inverting gain: Av = –Rf/Rin
Sine-wave slew-rate requirement: SRmin = 2πfVp
First-order closed-loop bandwidth estimate: fCL ≈ GBW / noise gain

12. PRESENTATION STORYBOARD (6 SLIDES)

Slide	Title	Content / visual
1	AI Datasheet Assistant	Project title, ECE team/student details, problem statement.
2	Why datasheets are difficult	Large PDFs, variant differences, tables, units, conditions and AI hallucination risk.
3	Proposed architecture	Flow diagram: PDFs → ingestion → retrieval → LLM → citations → verification.
4	Dataset and test plan	Ten op-amp families, 30 test questions, five task categories.
5	Demonstration and evaluation	Show one lookup, one calculation and one abstention; insert actual measured results.
6	Conclusion and future work	Benefits, limitations, human verification and planned improvements.

Suggested live demonstration

- Ask: “What is the GBW of the exact LM358 variant in the uploaded PDF?” Show the answer with revision and page.
- Ask a calculation: “Calculate the non-inverting gain for Rf=90 kΩ and Rg=10 kΩ.” Show the formula and result 10 V/V.
- Ask an unsupported question: “What is the current retail price in Chennai?” The assistant must abstain because the datasheet does not contain retail pricing.
- Open the cited PDF page and manually confirm the claim.

13. PRIVACY, COPYRIGHT, SAFETY AND LIMITATIONS

Copyright

Datasheets are copyrighted manufacturer documents. Use them for educational analysis under applicable institutional rules and the document's terms. Prefer linking to official PDFs rather than redistributing a large bundle. Include attribution, manufacturer, title, revision and source URL. Do not publish extracted tables or entire datasheets as a replacement for the originals unless permission or licence terms allow it.

Privacy and data handling

The source documents are generally public, but prompts, uploaded files and project logs may be processed by third-party AI services. Do not upload confidential schematics, customer designs, unpublished product data, credentials or personal information without authorization. Check each service's data controls and institutional policy. Retain only the minimum logs needed for assessment.

Engineering safety

An AI answer is not an engineering approval. Absolute maximum ratings, input differential limits, thermal limits, output current and stability must be reviewed by a qualified person. A wrong supply or pinout can damage hardware. For any physical build, use current-limited supplies, correct decoupling, safe test procedures and appropriate supervision.

Limitations

- PDF table extraction may scramble rows, columns, footnotes or superscripts.
- Typical performance graphs may not be searchable as text and require visual inspection.
- Different manufacturers' versions of the same generic part number may not share specifications.
- AI answers may be fluent but unsupported; citations can be incomplete or incorrect.
- Some specifications depend strongly on temperature, load, supply voltage and frequency.
- Pricing, stock, lifetime in a particular circuit and PCB-level noise are not datasheet-only questions.

14. CONCLUSION AND FUTURE SCOPE

The AI Datasheet Assistant provides a structured method for searching operational-amplifier datasheets while keeping the original manufacturer document as the authority. The key design principle is evidence-first answering: every

numeric claim should be tied to the exact part variant, document revision, page and test conditions. The 30-question suite tests direct lookup, comparison, calculations, unanswerable questions and ambiguity. A human verification stage remains essential, especially for electrical limits and circuit design decisions. The project can be extended with table-aware PDF extraction, automatic unit normalization, SPICE model linking, component selection filters, version-change detection and a web interface for multiple component families.

Expected project outcomes

- A curated manifest of ten official operational-amplifier datasheets.
- A searchable, source-grounded question-answering workflow.
- A 30-question test set and reproducible evaluation log.
- A manual verification report with error classification.
- A prompt library, automation workflow, quick-reference card and six-slide presentation.
- A final live demonstration showing correct citation and safe abstention.

15. REFERENCES

Manufacturer source pages and datasheets (accessed 30 October 2026). Always confirm the current PDF revision before use.

1. Texas Instruments, LM358 / LM358B manufacturer documentation. https://www.ti.com/lit/ds/symlink/lm358a.pdf. Dual general-purpose amplifier family; use the exact B/BA suffix and revision.

2. Texas Instruments, LM741 manufacturer documentation. https://www.ti.com/product/LM741. Single general-purpose bipolar op amp; datasheet linked from product page.

3. Texas Instruments, OP07 manufacturer documentation. https://www.ti.com/product/OP07. Precision, low-offset single op amp; verify exact grade and conditions.

4. STMicroelectronics, TL081 manufacturer documentation. https://www.st.com/en/amplifiers-and-comparators/tl081.html. JFET-input single op amp.

5. STMicroelectronics, TL071 manufacturer documentation. https://www.st.com/en/amplifiers-and-comparators/tl071.html. JFET-input single op amp.

6. STMicroelectronics, LM324 manufacturer documentation. https://www.st.com/content/st_com/en/products/amplifiers-and-comparators/operational-amplifiers-op-amps/standard-op-amps/lm324.html. Quad general-purpose amplifier family.

7. Microchip Technology, MCP6002 manufacturer documentation. https://www.microchip.com/en-us/product/MCP6002. Dual, low-power 1 MHz op amp family.

8. Texas Instruments, NE5532 manufacturer documentation. https://www.ti.com/product/NE5532. Dual low-noise audio op amp.

9. Texas Instruments, OPA2134 manufacturer documentation. https://www.ti.com/product/OPA2134. Dual audio op amp; FET input.

10. Analog Devices, AD822 manufacturer documentation. https://www.analog.com/en/products/ad822.html. Dual, single-supply FET-input amplifier.

Additional technical reference

Texas Instruments, “Understanding Operational Amplifier Specifications” and the relevant device datasheets. Use manufacturer application notes for definitions and design context, but use the selected device datasheet for guaranteed electrical limits.

APPENDIX A — FINAL SUBMISSION CHECKLIST

- Ten official PDFs saved and manifest completed with revision/date/page count.
- 30 test questions run in at least two AI tools; transcripts/screenshots retained.
- Prompt comparison completed using identical questions and source documents.
- Every numeric answer manually checked against original PDF.
- Accuracy, citation precision, unit error and hallucination metrics calculated from real audit records.
- Quick-reference card and 5–7 slide presentation exported.
- Automation workflow demonstrated and output log attached.
- Copyright, privacy, AI limitations and engineering safety discussed.

• Student name, register number, institution and guide details filled on cover.

APPENDIX B — SAMPLE FINAL ANSWER FORMAT

Part: [exact manufacturer + full part number]
Parameter: [requested parameter]
Answer: [value + unit]
Statistic: [minimum / typical / maximum]
Conditions: [supply, load, temperature, frequency]
Source: [document title, revision, page, section]
Evidence: [short excerpt or table row]
Verification: [PASS / FAIL / NEEDS REVIEW]
Note: [limitations or assumptions]

END OF PROJECT REPORT